Exploratory Data Analysis of Airline Delays and Cancellations

Name: Ahmadreza Jafari **Student ID:** 403222080 **Course:** Data Science —Shahid Beheshti University, Spring 2026 **Instructor:** Dr. Kheradpisheh

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1. Introduction

Flight delays and cancellations are a routine feature of commercial air travel and a well-studied problem in applied statistics and operations research, since airline systems log detailed, high-volume operational data over long periods. This report presents an exploratory data analysis (EDA) of a subset (2015–2017) of the *Airline Delay and Cancellation Data, 2009–2018* dataset published on Kaggle by Yuanyu Wendy Mu, which records US domestic flights with information on carriers, airports, scheduled and actual timings, delay measures, cancellations, diversions, and delay-cause breakdowns.

The aim of this analysis is descriptive rather than causal: to characterise the structure and quality of the data, identify the main patterns in delays and cancellations over time, across carriers, and across airports, and to examine relationships between the numeric variables that describe flight timing and delay. These patterns are useful both as a standalone description of airline performance in this period and as groundwork for any later predictive modelling.

2. Data and methods

2.1 Data source

The data come from the Kaggle dataset *Airline Delay and Cancellation Data, 2009–2018*, distributed as one CSV file per year. This analysis uses the **2015, 2016, and 2017** files, concatenated into a single dataframe of **17,111,358 rows and 28 columns** (27 from the source files plus a derived `YEAR` column). For performance and memory efficiency, the yearly CSVs were converted once to Parquet (keeping only the 18 columns the analysis uses plus `YEAR`, with float columns stored as `float32`), which reduces load time from ~45 s to under 2 s; the notebook falls back to the CSVs if the Parquet files are absent.

2.2 Key variables

Field	Category	Description
`FL_DATE`	Identifier / time	Flight date
`OP_CARRIER`	Identifier	Operating carrier code (2-letter)
`ORIGIN`, `DEST`	Identifier	Origin / destination airport code
`CRS_DEP_TIME`	Time	Scheduled departure time (local, HHMM)
`DEP_DELAY`, `ARR_DELAY`	Delay measure	Departure / arrival delay in minutes (negative = early)
`TAXI_OUT`, `TAXI_IN`	Time	Taxi-out / taxi-in duration in minutes
`CANCELLED`, `CANCELLATION_CODE`	Indicator	Cancellation flag and reason code (A/B/C/D)
`DIVERTED`	Indicator	Diversion flag
`DISTANCE`	Numeric	Great-circle distance in miles

Field	Category	Description
CARRIER_DELAY, WEATHER_DELAY, NAS_DELAY, SECURITY_DELAY, LATE_AIRCRAFT_DELAY	Delay measure	Minutes of delay attributed to each cause (only populated when ARR_DELAY ≥ 15)

2.3 Preprocessing

FL_DATE was parsed as a datetime64 column; OP_CARRIER, ORIGIN, DEST, and CANCELLATION_CODE were cast to category dtype for memory efficiency. A YEAR column was added at load time.

2.4 Missing data

The proportion of missing values for the main variables is:

Column	Missing	% missing
CANCELLATION_CODE	16,872,920	98.61%
SECURITY_DELAY	14,054,207	82.13%
CARRIER_DELAY	14,054,207	82.13%
NAS_DELAY	14,054,207	82.13%
WEATHER_DELAY	14,054,207	82.13%
LATE_AIRCRAFT_DELAY	14,054,207	82.13%
ARR_DELAY	279,795	1.64%
TAXI_IN	245,031	1.43%
TAXI_OUT	236,610	1.38%
DEP_DELAY	229,952	1.34%

The five delay-cause columns (CARRIER_DELAY, WEATHER_DELAY, NAS_DELAY, SECURITY_DELAY, LATE_AIRCRAFT_DELAY) show identical, very high missingness (82.13%). Per the dataset documentation, these are only populated by the reporting carrier when a flight's arrival delay is at least 15 minutes; for flights delayed less than that (or not delayed / cancelled), the fields are legitimately absent rather than "missing"in the data-quality sense. We treat NaN in these columns as "no delay of this cause / not applicable", **not** as a value to impute.

DEP_TIME, ARR_TIME, DEP_DELAY, ARR_DELAY, TAXI_OUT, TAXI_IN are missing almost exclusively for **cancelled** flights, since a cancelled flight never actually departs or arrives. This was verified by cross-tabulating missingness against the CANCELLED indicator: **100.0% of missing DEP_DELAY rows and 85.2% of missing ARR_DELAY rows correspond to cancelled flights** (the remaining ~15% of missing ARR_DELAY are diverted flights, which depart but do not arrive at the scheduled destination).

Rows with missing delay values were **kept** when counting flights (by carrier, airport, time period) but **dropped** when analysing the distribution of delay itself, since a cancelled flight has no realised delay value.

2.5 Derived variables

- ARR_DEL15 —binary indicator for arrival delay of at least 15 minutes (missing for cancelled flights).

- `DEP_HOUR` and `TIME_OF_DAY` —scheduled departure hour (0–23) and a four-level time-of-day bucket (Morning 6–11 / Afternoon 12–17 / Evening 18–23 / Night 0–5).
- `DAY_OF_WEEK` —day-of-week name derived from `FL_DATE`.
- `TOTAL_DELAY_CAUSE_MIN` —sum of the five delay-cause columns (`NaN` treated as 0).
- `MAIN_DELAY_CAUSE` —the dominant delay cause per delayed flight (only defined where at least one cause column is populated).

2.6 Tools

Analysis was carried out in Python using `pandas` 3.0 for data manipulation and `matplotlib` / `seaborn` for visualisation, in a Jupyter notebook (`EDA_Airline_Delays_403222080.ipynb`) designed to run top-to-bottom without manual intervention.

3. Exploratory analysis

3.1 Data overview

The dataset spans **17,111,358 flights** across the three years: 5,819,079 in 2015, 5,617,658 in 2016, and 5,674,621 in 2017. `describe()` on the numeric delay columns shows means well above their medians —mean `ARR_DELAY` = 4.09 min vs. median = −6 min; mean `DEP_DELAY` = 9.35 min vs. median = −2 min — consistent with a right-skewed distribution driven by a long tail of large delays (the maximum arrival delay is 2,189 min, over 36 hours). **1.39%** of flights were cancelled and **0.24%** were diverted overall.

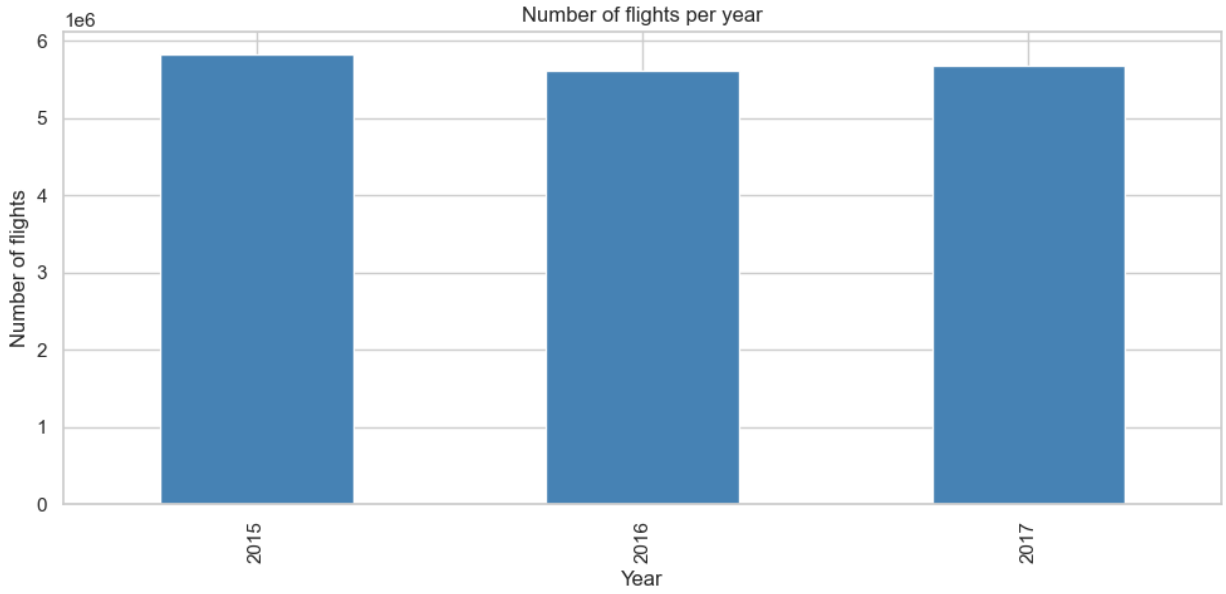


Figure 1: Flights per year

Figure 1. Number of flights per year. Volume is broadly stable, with a slight dip in 2016.

Figure 2. Number of flights per month (2015–2017 combined). Summer months (Jun–Aug) and March are the busiest; February is the quietest.

Figure 3. Distributions of arrival delay (left) and departure delay (right), clipped to [−50, 200] min for visualisation. Both are sharply peaked just below zero —most flights depart and arrive close to schedule, or

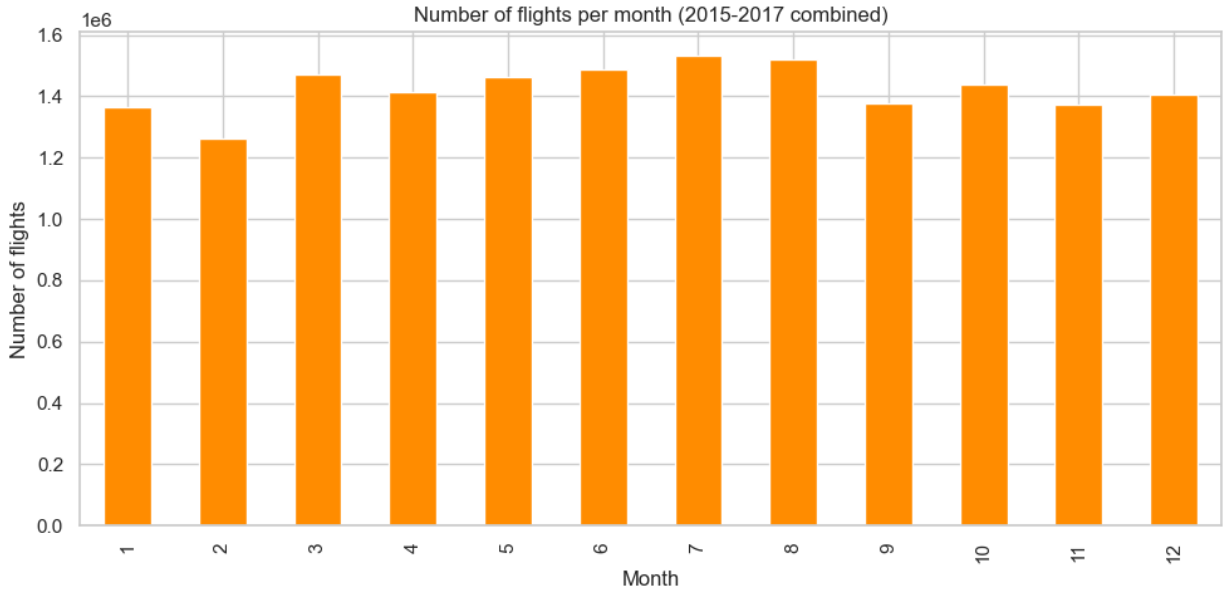


Figure 2: Flights per month

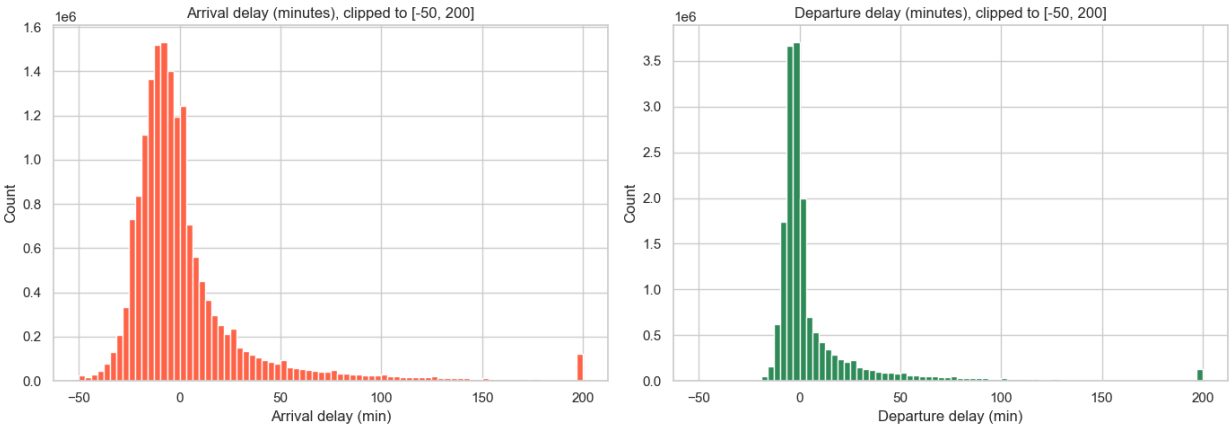


Figure 3: Delay histograms

slightly early —with a long right tail. The clipping is visual only; the underlying data contain delays up to 2,755 min (departure) and 2,189 min (arrival).

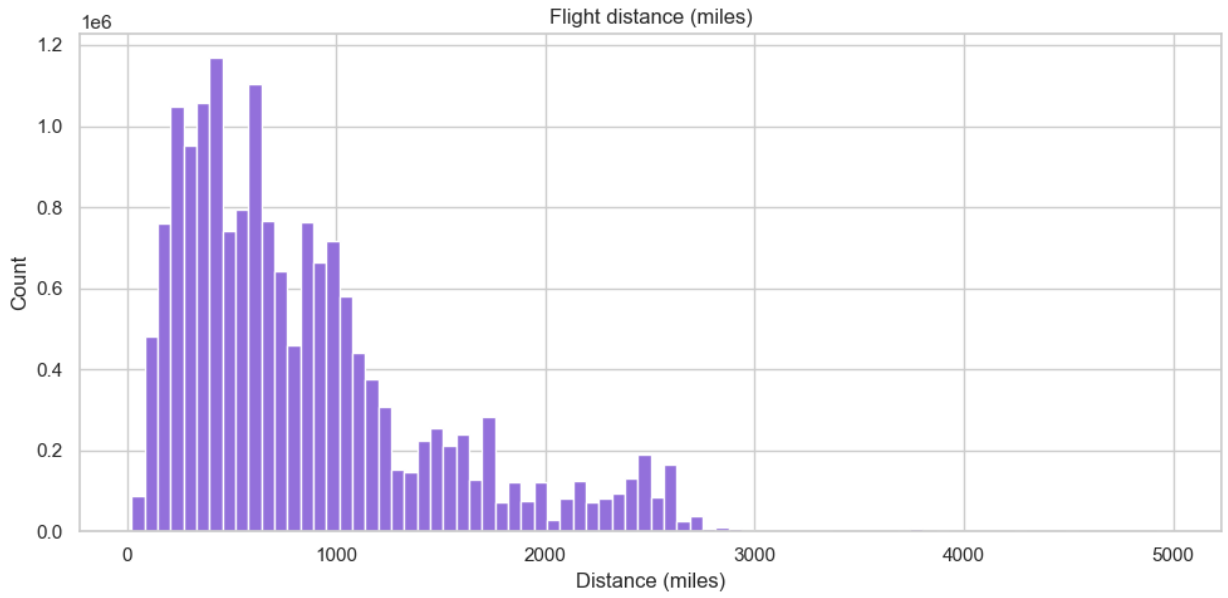


Figure 4: Distance histogram

Figure 4. Distribution of flight distance. The distribution is right-skewed and roughly multimodal, reflecting common route-length bands; the bulk of flights are under ~1,100 miles (75th percentile), with a long tail of long-haul flights up to 4,983 miles.

Figure 5. Top 10 carriers by number of flights. Southwest (WN) dominates with 3.89M flights, followed by Delta (DL, 2.72M), American (AA, 2.54M), SkyWest (OO, 1.90M), and United (UA, 1.65M).

Figure 6. Top 10 origin airports by departures. Atlanta (ATL, 1.13M departures) is by far the busiest, followed by Chicago O'Hare (ORD), Denver (DEN), Los Angeles (LAX), and Dallas/Fort Worth (DFW) — the major US hubs.

3.2 Time patterns

Figure 7. Average arrival delay by month, 2015–2017. Delays peak in summer (Jun–Aug) and again in December/January, consistent with higher traffic volume and convective-weather disruption in summer and winter weather in midwinter.

Figure 8. Number of flights by month, 2015–2017. Volume tracks the seasonal pattern of Figure 2 over each year.

Figure 9. Average arrival delay by year. 2016 (3.52 min) was a noticeably better year than 2015 (4.41 min) or 2017 (4.33 min), but the levels are broadly similar across the three years.

*Figure 10. Cancellation rate (%) by month, 2015–2017. Cancellations show an even sharper seasonal pattern than delays: the single worst month is **February 2015 at 4.78% cancelled** (the well-documented severe winter of early 2015), while the best is November 2016 at 0.29%.*

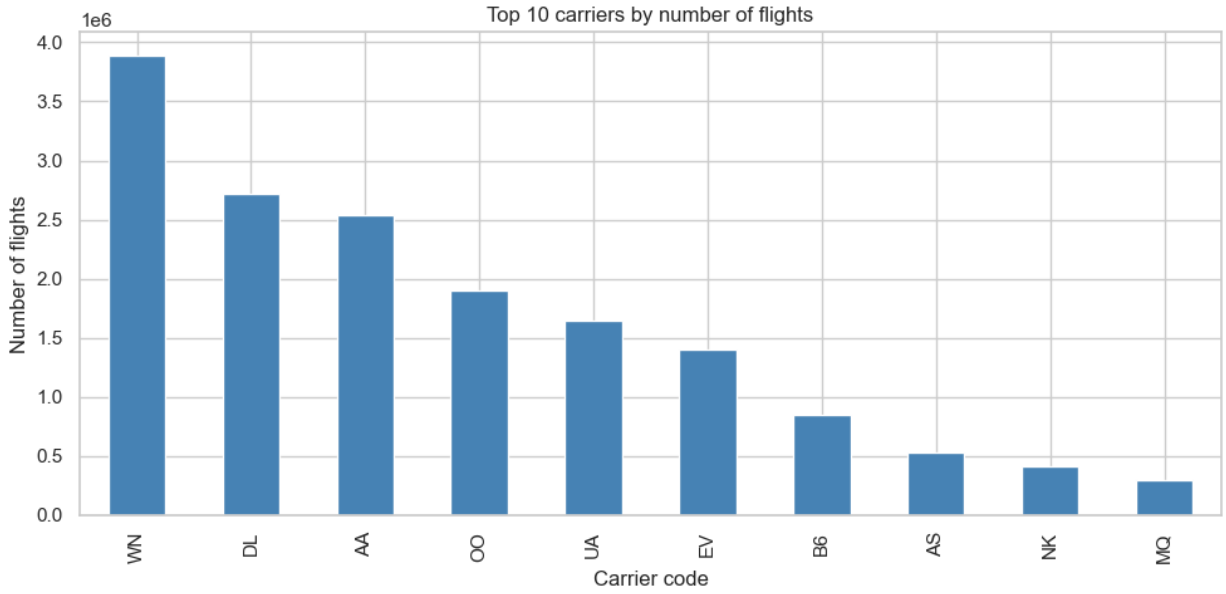


Figure 5: Top carriers

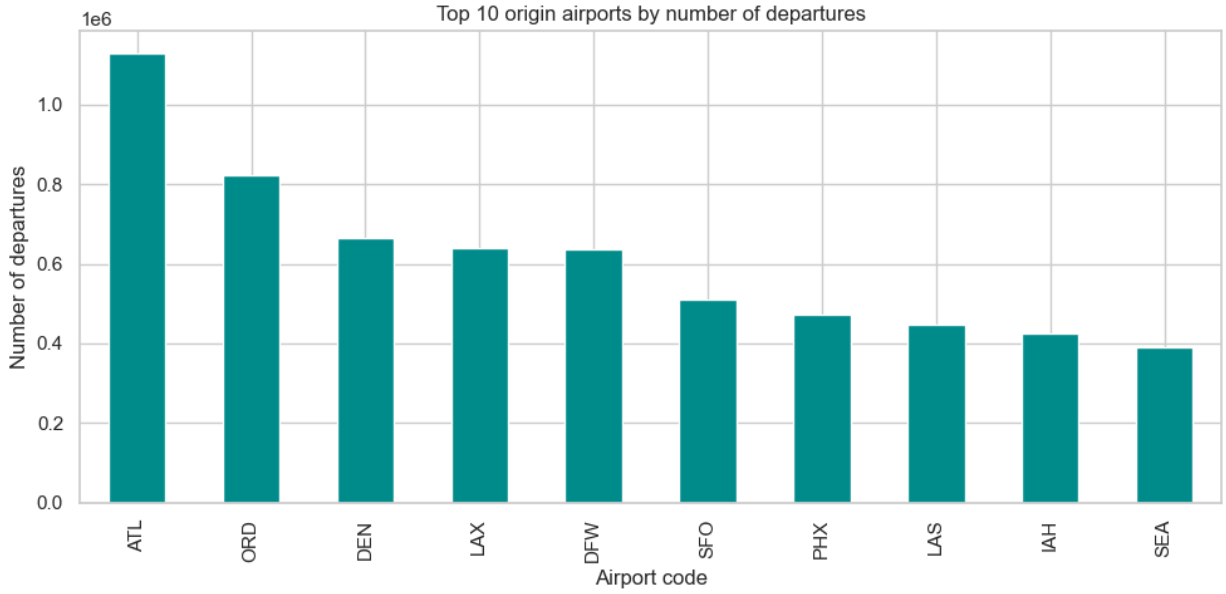


Figure 6: Top origins

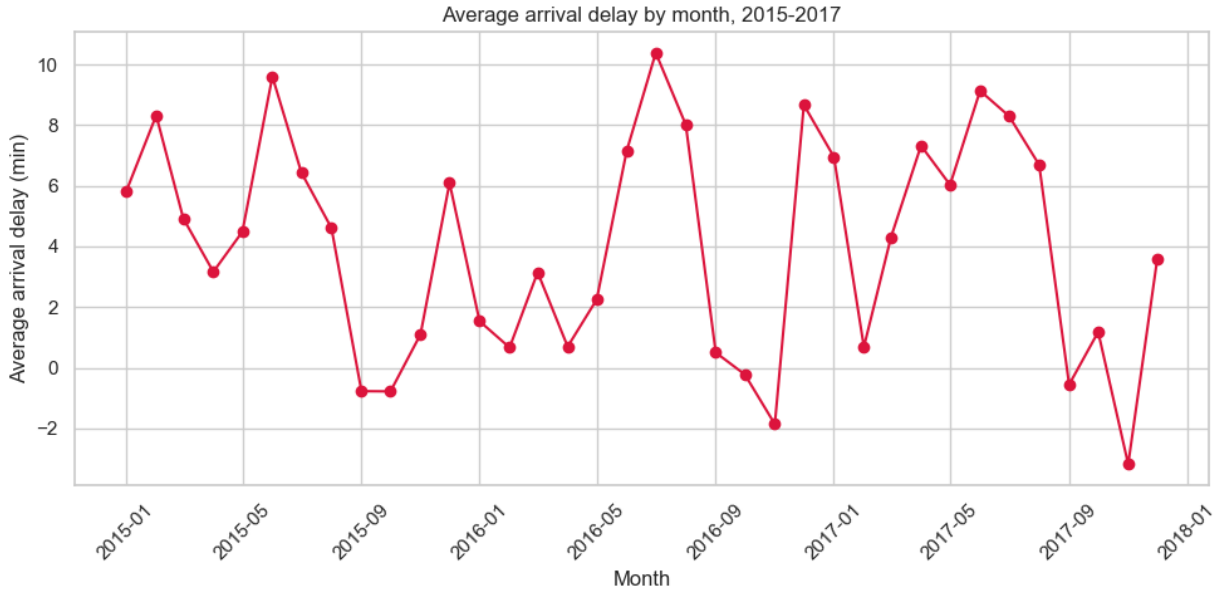


Figure 7: Average delay by month

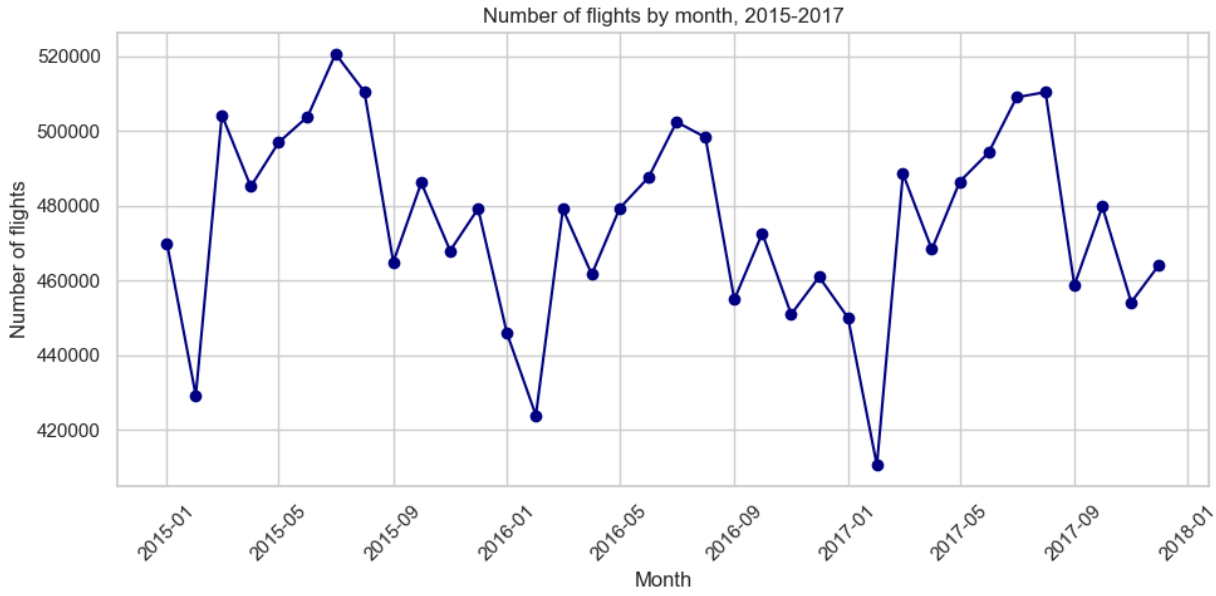


Figure 8: Flights by month

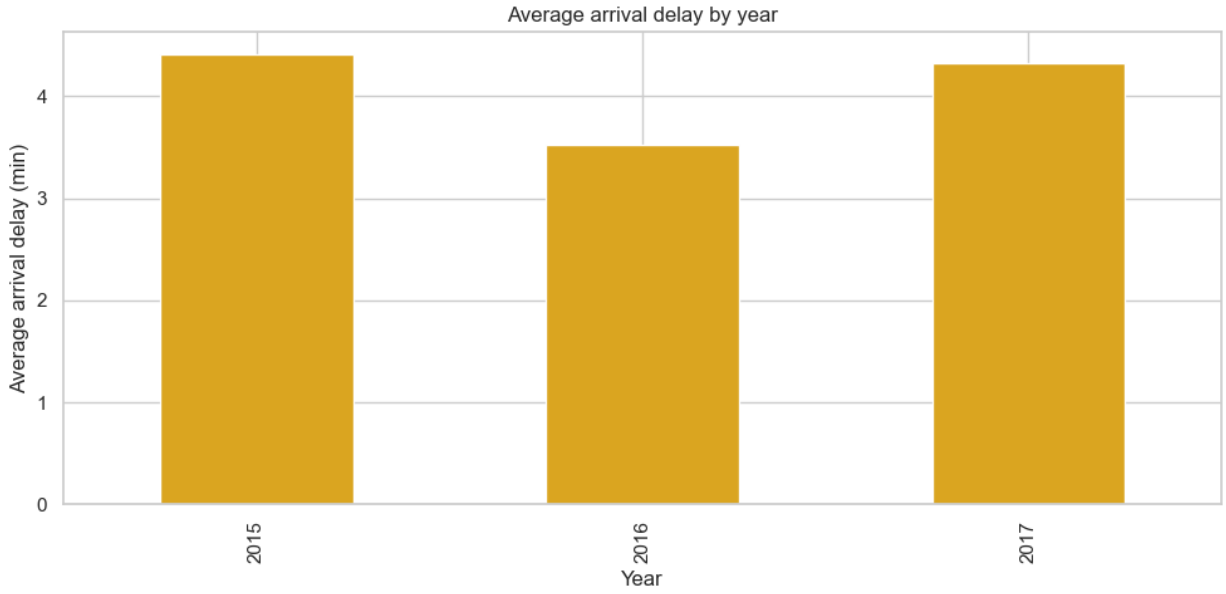


Figure 9: Average delay by year

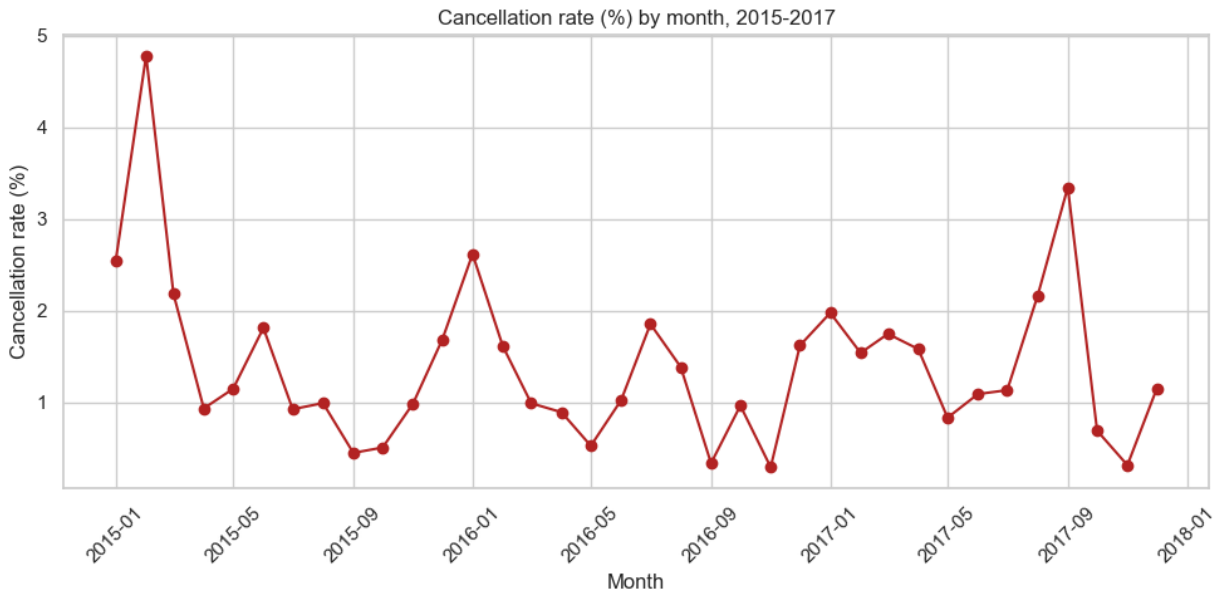


Figure 10: Cancellation rate by month

3.3 Day of week and time of day

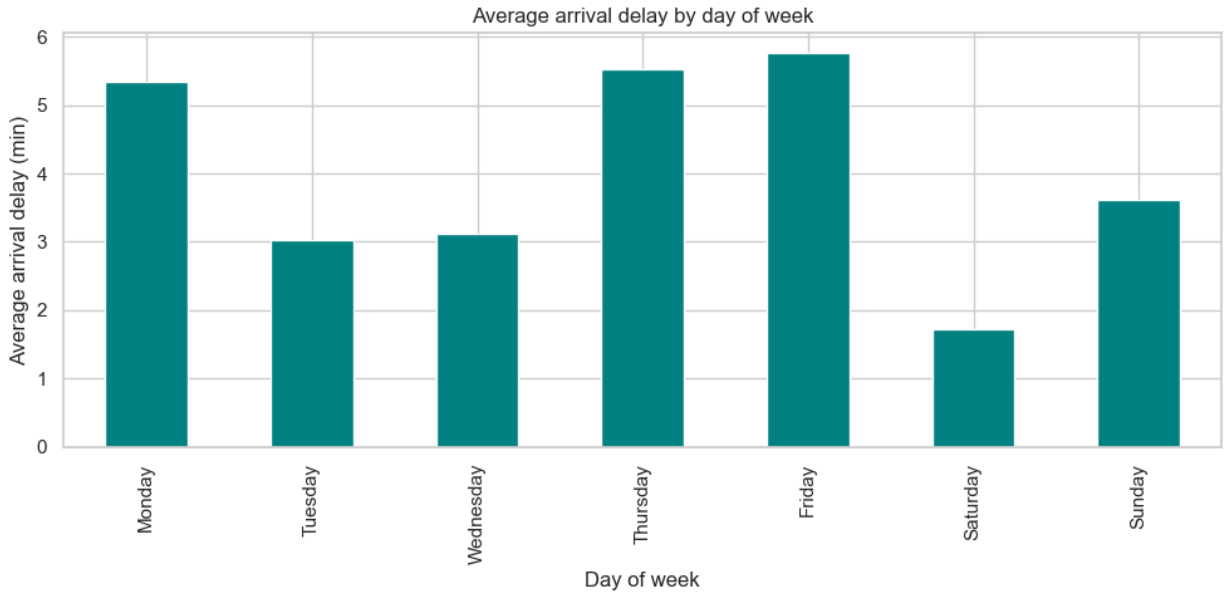


Figure 11: Delay by day of week

Figure 11. Average arrival delay by day of week. Saturday (+1.73 min) is the best day; Friday (+5.76 min), Thursday (+5.53 min), and Monday (+5.34 min) are the worst, consistent with business- travel peaks on Monday morning and Thursday/Friday evening and lighter schedules on Saturday.

Figure 12. Average arrival delay by time-of-day bucket (scheduled departure). Delays build up over the course of the day: Morning flights (06–11) average −0.36 min (essentially on time), Afternoon (12–17) +6.39 min, Evening (18–23) +9.01 min, and Night (00–05) −3.30 min. This is the classic operational knock-on pattern —delays accumulated on earlier rotations propagate through the day. We note this is purely descriptive/associational, not a causal claim.

3.4 Carriers, airports, and routes

Figure 13. Average arrival delay by carrier. Alaska (AS, −0.83 min), Delta (DL, −0.15 min), and Hawaiian (HA, +1.29 min) lead; Spirit (NK, +9.39 min), JetBlue (B6, +8.64 min), and Frontier (F9, +8.52 min) are the worst.

Figure 14. Cancellation rate (%) by carrier. Hawaiian (HA, 0.22%), Alaska (AS, 0.56%), and Delta (DL, 0.62%) lead on cancellations too; Envoy/Mesa (MQ, 5.10%), ExpressJet (EV, 2.64%), and Spirit (NK, 2.37%) are the worst.

The carriers that do well on delay also tend to do well on cancellation (AS, DL, HA at the top of both rankings), and the low-cost carriers NK/B6/F9 do poorly on delay. MQ and EV (regional carriers) have disproportionately high cancellation rates despite middling average delays —a different operational story. The full ranking is:

Carrier	Avg arrival delay (min)	Cancel rate (%)	N flights
AS	−0.83	0.559	534,869

Carrier	Avg arrival delay (min)	Cancel rate (%)	N flights
DL	−0.15	0.621	2,722,187
HA	+1.29	0.223	233,233
UA	+2.90	1.094	1,645,271
US	+3.71	2.047	198,715
AA	+4.18	1.337	2,536,827
WN	+4.45	1.284	3,890,743
OO	+5.80	1.519	1,900,813
MQ	+6.46	5.100	294,632
EV	+6.53	2.644	1,402,508
VX	+7.88	1.107	202,005
F9	+8.52	0.946	288,984
B6	+8.64	1.929	848,175
NK	+9.39	2.370	412,396

Figure 15. Average arrival delay at the 15 busiest origin airports. MSP (+0.72 min), SEA (+1.91 min), and ATL (+2.69 min) perform best; SFO (+7.18 min), ORD (+7.13 min), and DFW (+7.00 min) perform worst among the busiest origins.

Figure 16. Average arrival delay at the 15 busiest destination airports. ATL (+1.20 min), MSP (+1.55 min), and SEA (+1.70 min) perform best; EWR (+10.41 min), SFO (+10.04 min), and BOS (+6.49 min) perform worst. Note that EWR and SFO appear at the worse end of both origin and destination rankings —the New York and San Francisco metropolitan areas are chronically congested.

The top 10 routes by flight count are predominantly California short-hauls (SFO↔LAX, LAX↔LAS, LAX↔SEA) plus the transcon LAX↔JFK and the Chicago–New York LGA↔ORD shuttle. The California routes show notably high average arrival delays (SFO→LAX +13.59 min, LAX→SFO +15.20 min), while the transcon JFK↔LAX routes are essentially on time (−0.14 min and +3.65 min).

Route	N flights	Avg arrival delay (min)
SFO-LAX	48,585	+13.59
LAX-SFO	47,680	+15.20
JFK-LAX	38,434	−0.14
LAX-JFK	38,424	+3.65
LAS-LAX	33,166	+11.95
LAX-LAS	33,068	+9.14
LGA-ORD	30,898	+2.19
ORD-LGA	30,586	+9.67
SEA-LAX	29,024	+5.01
LAX-SEA	28,960	+2.83

3.5 Relationships between variables

*Figure 17. Departure delay vs. arrival delay (hexbin, 50,000-flight random sample). Departure and arrival delay are **very strongly correlated (Pearson r = 0.954)**, consistent with the expectation that a flight leaving late tends to arrive late by a similar margin. The hexbin forms a tight diagonal band with some spread above*

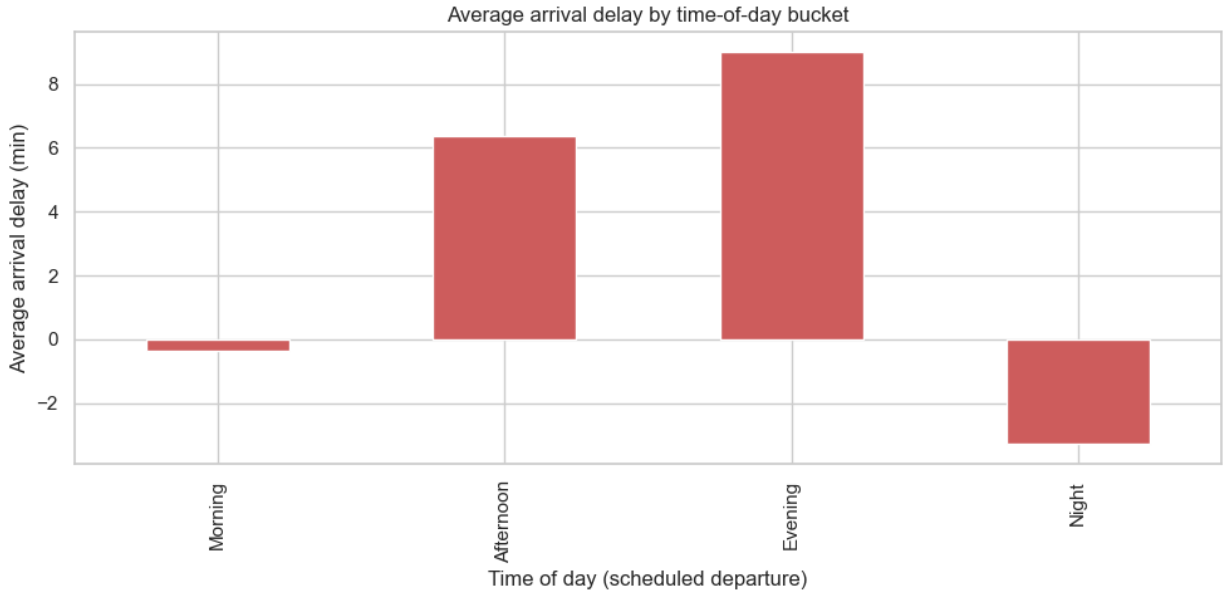


Figure 12: Delay by time of day

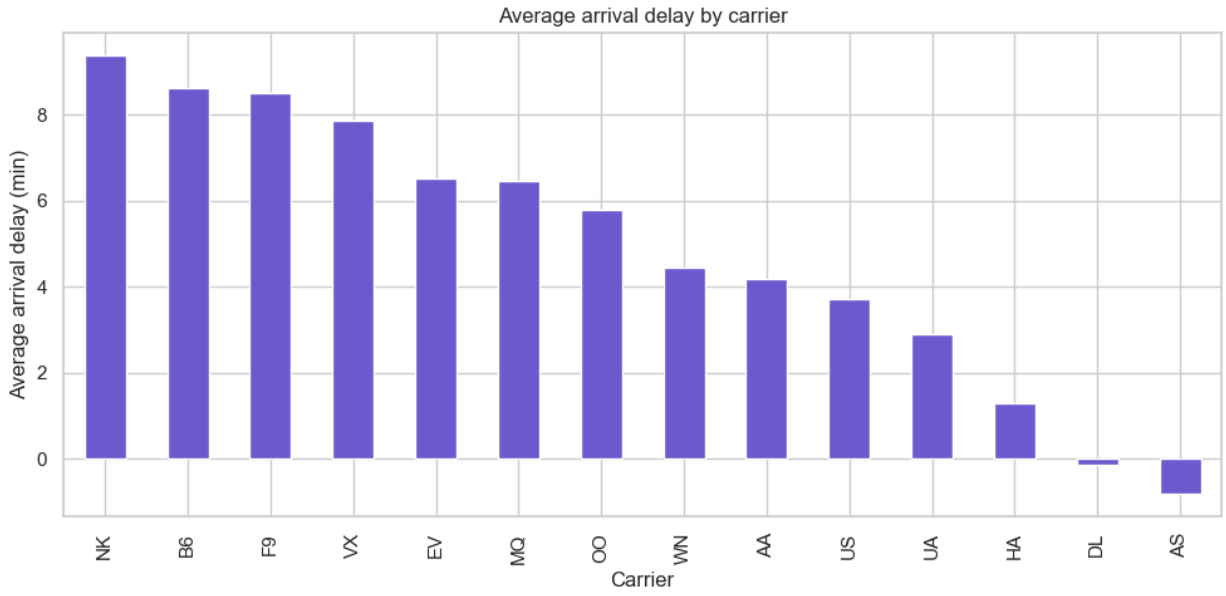


Figure 13: Delay by carrier

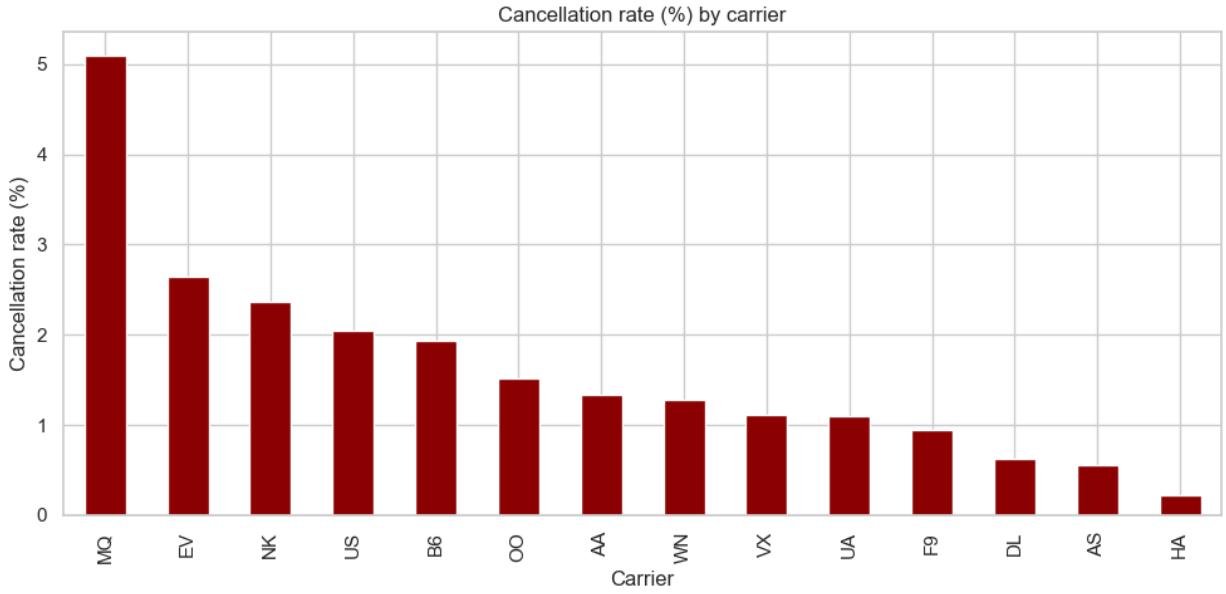


Figure 14: Cancellation rate by carrier

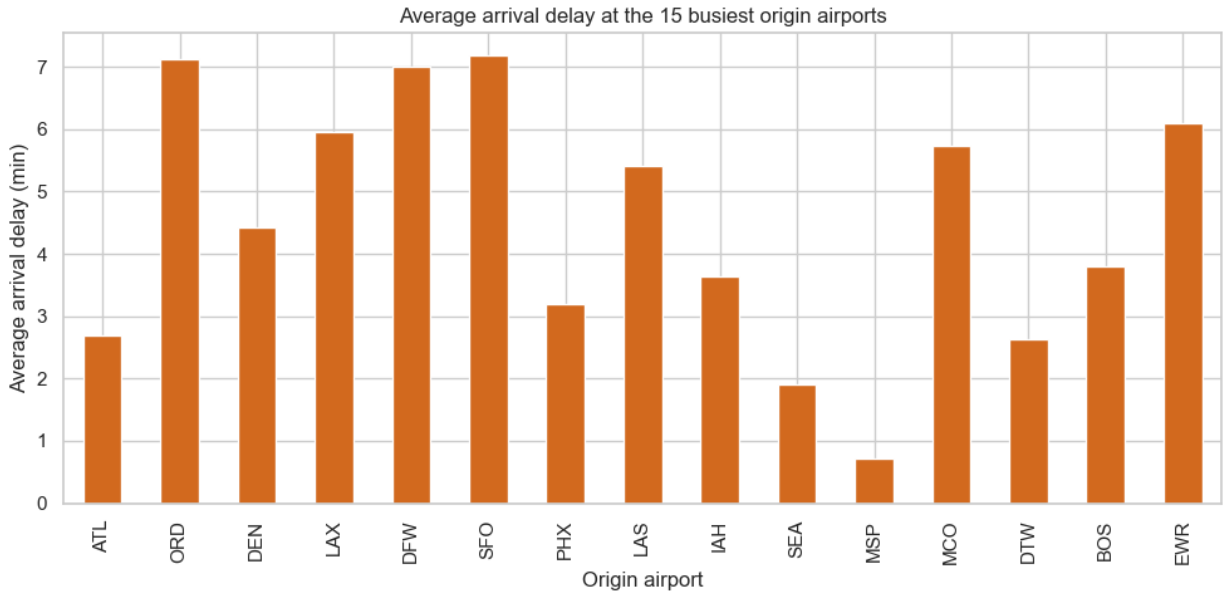


Figure 15: Origin delay

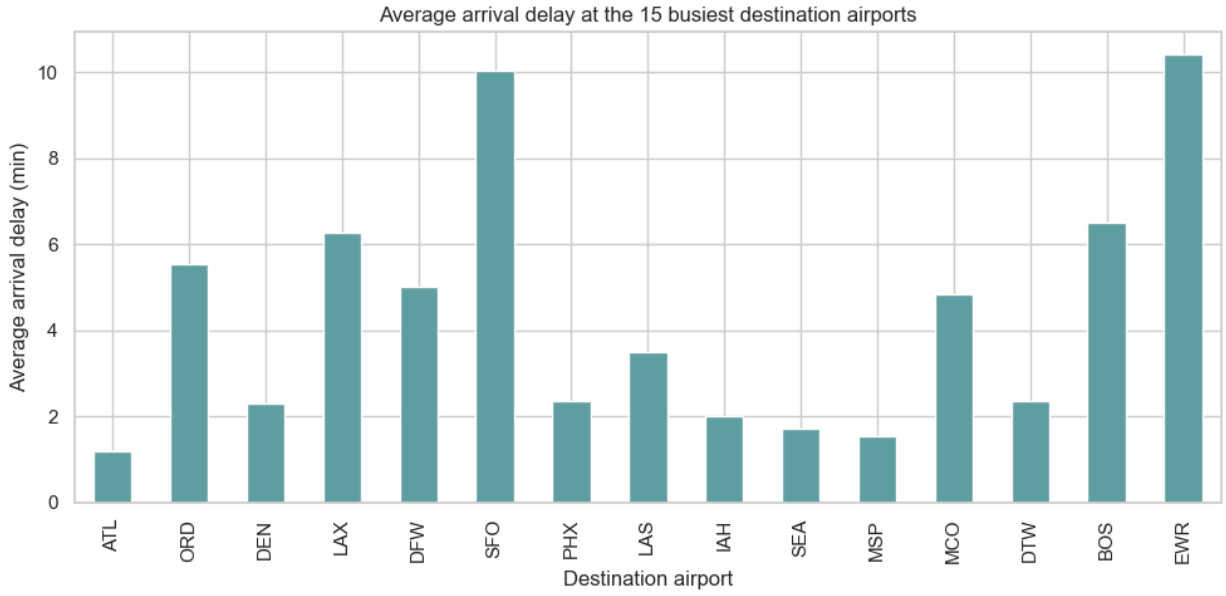


Figure 16: Destination delay

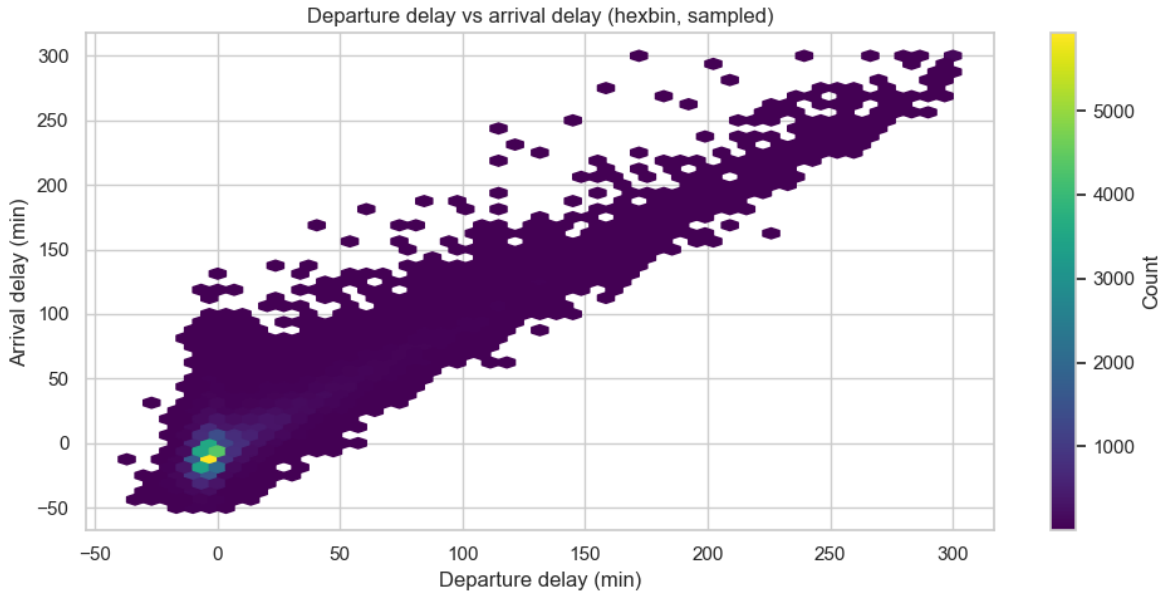


Figure 17: Departure vs arrival delay

the diagonal (flights that lose a little more time in the air) and below it (flights that make up some of the departure delay en route).

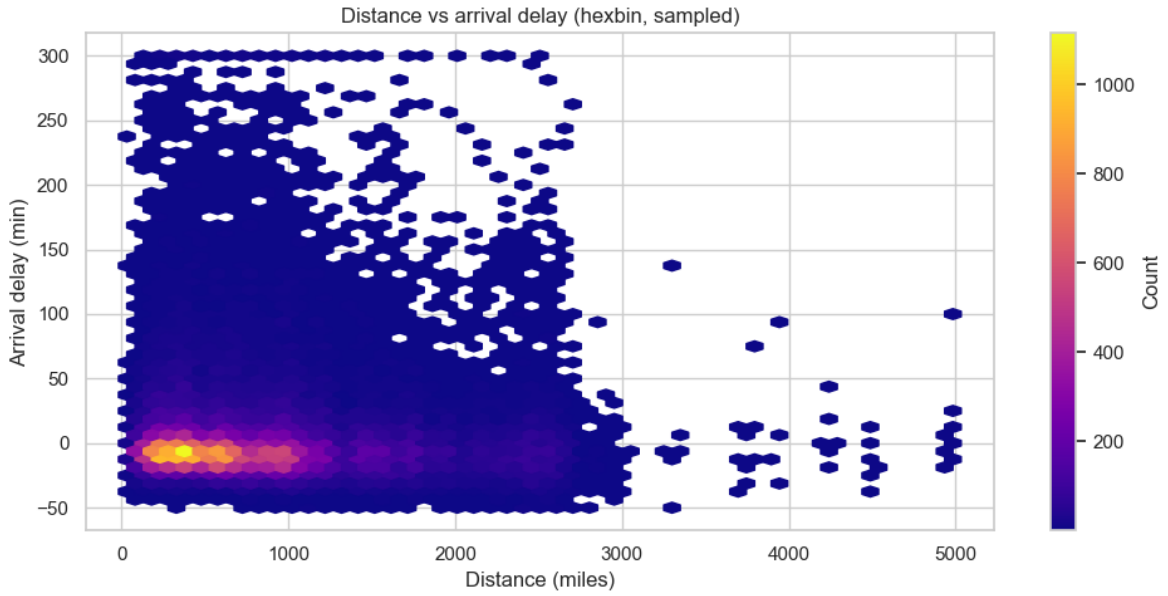


Figure 18: Distance vs arrival delay

Figure 18. Distance vs. arrival delay (hexbin, 50,000-flight random sample). The relationship between distance and arrival delay is essentially ***flat (r = −0.011)****. Flight length alone is not a meaningful driver of delay; both short-haul and long-haul flights show the same wide vertical spread in arrival delay.*

Figure 19. Correlation matrix of delay-related numeric variables (computed on the full 17M-row dataset). The strongest correlation is `DEP_DELAY` ↔ `ARR_DELAY` *(r = 0.95).* `ARR_DELAY` *is next most correlated with the operational cause columns* `CARRIER_DELAY` *(0.66) and* `LATE_AIRCRAFT_DELAY` *(0.49).* `DISTANCE` *is essentially uncorrelated with every other variable (|r| ≤ 0.08). The five delay-cause columns are roughly independent of one another (|r| ≤ 0.15), which makes sense as they are defined as additive components of a flight's total delay.*

Figure 20. Share of total delay minutes by cause, among the 3,057,152 flights delayed by at least 15 minutes. ***Late-aircraft delay dominates at 39.5%****, followed by carrier delay (32.0%), NAS delay (23.9%), weather delay (4.5%), and security delay (0.1%).*

This suggests that delay in 2015–2017 is driven more by airline operational factors and knock-on late-aircraft effects than by weather or security events, at least in aggregate —weather and security together account for under 5% of total delay minutes despite receiving disproportionate public attention. The total minutes attributed to each cause:

Delay cause	Total minutes	Share (%)
LATE_AIRCRAFT_DELAY	74,325,392	39.47
CARRIER_DELAY	60,222,996	31.98
NAS_DELAY	45,007,168	23.90
WEATHER_DELAY	8,506,075	4.52
SECURITY_DELAY	257,080	0.14

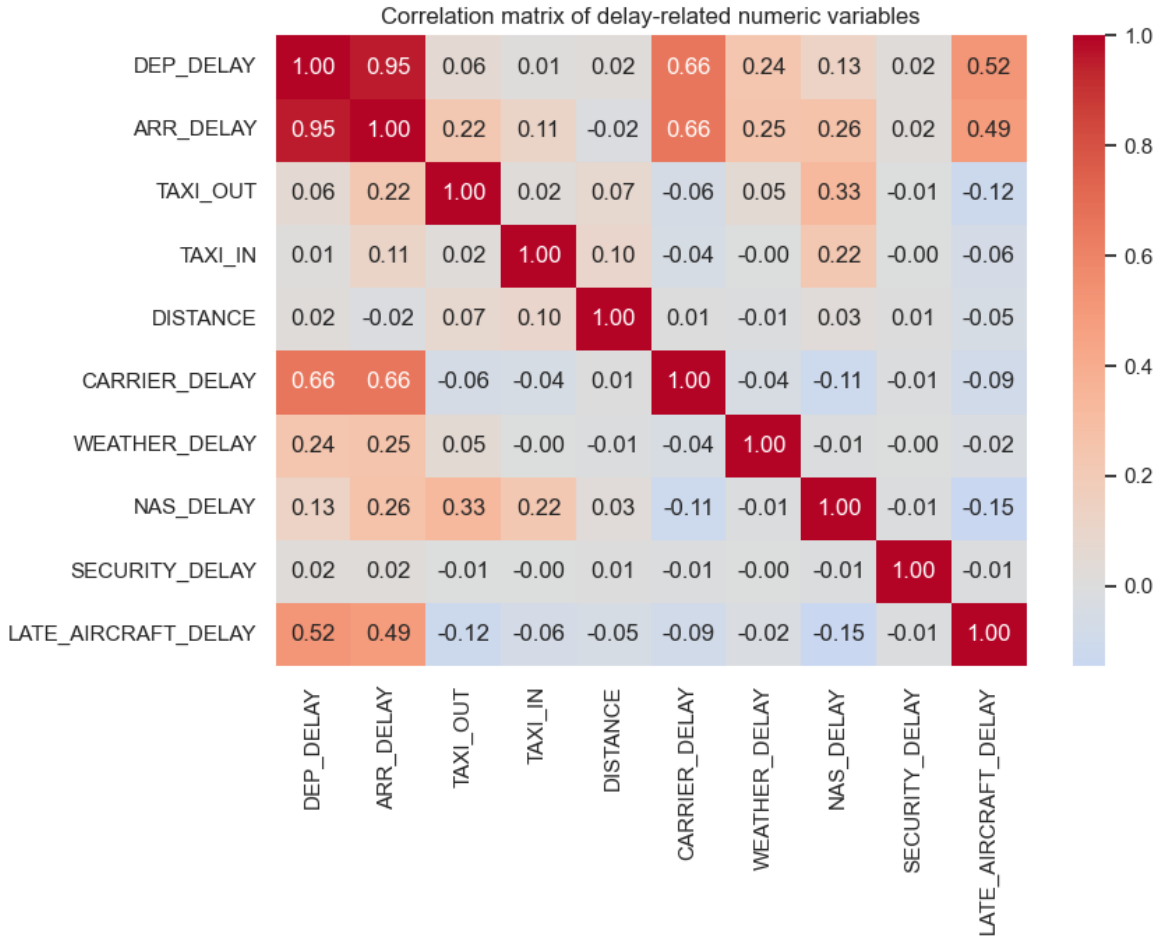


Figure 19: Correlation heatmap

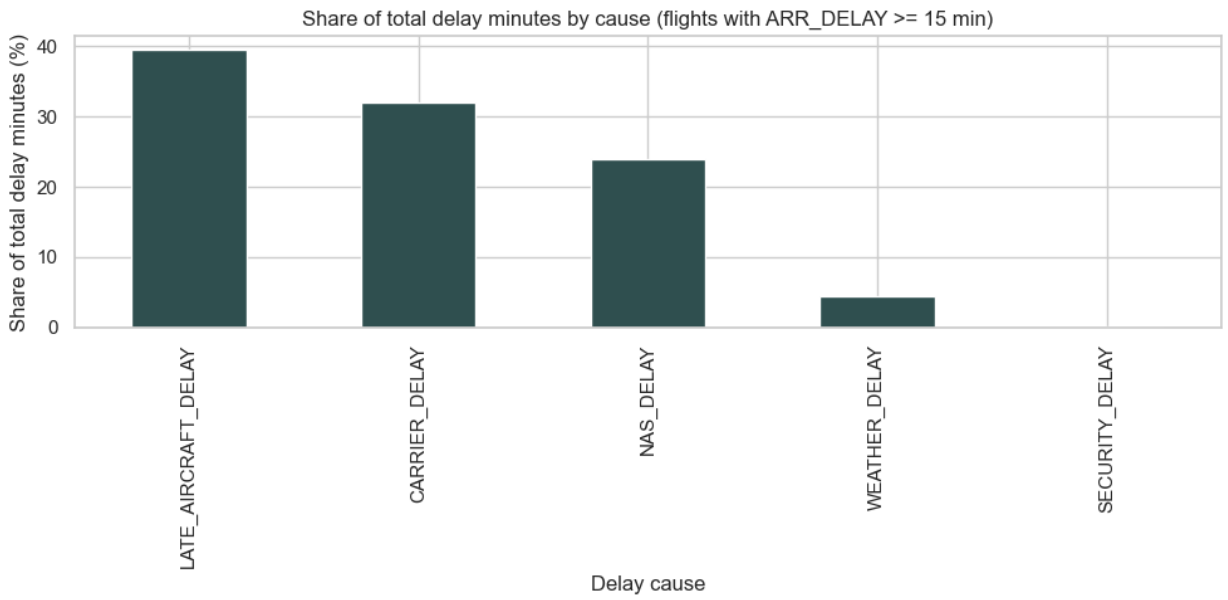


Figure 20: Delay cause share

4. Discussion and limitations

The EDA reveals a US domestic air-travel system in which the typical flight actually runs a few minutes early (median arrival delay −6 min), but whose positive mean delay (4.1 min) is driven by a long right tail of large delays. Performance varies systematically along several axes: by **season** (winter and summer peaks in both delay and cancellation, with February 2015 standing out as a severe-weather outlier), by **time of day** (delays accumulate through the day from −0.4 min in the morning to +9.0 min in the evening, consistent with knock-on propagation across rotations), by **day of week** (Saturday the best, Thursday/Friday/Monday the worst), by **carrier** (Alaska, Delta, and Hawaiian consistently best; Spirit, JetBlue, and Frontier worst on delay; regional carriers MQ and EV worst on cancellation), and by **airport** (the New York and San Francisco metro areas —EWR, SFO —chronically congested at both ends). The dominant delay causes are late-aircraft and carrier delay (together ~72% of all delay minutes among delayed flights), with weather and security contributing under 5%.

The strongest pairwise relationship in the data is between departure delay and arrival delay ($r = 0.95$), which is both expected and useful —it means pre-departure delay is an excellent proxy for the eventual arrival delay a passenger experiences. Distance, by contrast, is essentially uncorrelated with delay ($r \approx 0$), so flight length alone is not a useful predictor of delay.

Limitations of this analysis include: the dataset covers only three of the ten available years and may not generalise to the full 2009–2018 period or to more recent years; delay-cause attribution is self-reported by carriers and may vary in consistency across airlines; the analysis is purely descriptive/associational and does not control for confounding factors (airport congestion, aircraft type, weather severity, time of day) that would be needed to draw causal conclusions about *why* certain carriers or airports perform differently; and extreme delay values were clipped only for visualisation, not removed from the underlying statistics, so summary means remain sensitive to a small number of very large delays.

5. Bonus work

To make the analysis tractable on a 16 GB machine loading ~17 M rows, the three yearly CSV files (~2 GB total) were converted once to Parquet, keeping only the 18 columns the analysis uses and storing float columns as `float32`. This reduced load time from ~45 s (CSV) to under 2 s (Parquet) and cut peak memory enough to fit comfortably. The notebook documents both paths and falls back to CSV if Parquet is absent, so it remains reproducible from the original Kaggle files alone.

6. Conclusion

The 2015–2017 US domestic flight data describe a system dominated by a handful of large carriers and hub airports, in which most flights run close to or ahead of schedule but a long tail of large delays produces a positive mean. Delay and cancellation are strongly seasonal (peaking in winter and summer), accumulate through the day, and are lowest on Saturdays. Carriers separate clearly into better (Alaska, Delta, Hawaiian) and worse (Spirit, JetBlue, Frontier; regional carriers MQ and EV on cancellation) performers, and the New York/San Francisco metro airports are chronically congested. Among delayed flights, late-aircraft and carrier delay dominate; weather and security are minor in aggregate. Departure delay is an excellent proxy for arrival delay ($r = 0.95$), while distance is not. A natural next step would be to build a predictive model for the binary "arrival delay $\geq$ 15 min"outcome using pre-departure features (carrier, origin, scheduled hour, day of week, distance, month), or to extend the analysis to the full 2009–2018 period.

7. References

1. Mu, Y. W. *Airline Delay and Cancellation Data, 2009–2018*. Kaggle. https://www.kaggle.com/datasets/yuanyuwendyn
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